

MATH 2228-601, SUMMER II 2026, MOCK MIDTERM 3

INSTRUCTIONS

1. Do not open the exam paper or start working until you are told to do so.
2. Bubble in your name as DOE John S (if it is John S. Doe), bubble in your student ID number.
3. NO NOTES OR BOOKS ARE ALLOWED. You may not communicate with others during the exam. Please be considerate and refrain from making noise. Switch off any beepers, cell phones, etc.
4. Bubble in your answer to, say, Question 4, on line 4.
5. The exam is self-explanatory. The instructor will not interpret the questions for you. You may, however, ask for another copy of the exam if your copy is illegible.
6. Have your calculator in **RADIAN MODE**. You can set radian mode by pressing a button called MODE, SET UP, or DRG (degree, radian, gradian).

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1. Let $X_1, X_2, \dots, X_{19}$ be independent normally distributed random variables with mean -28 and standard deviation 1.4 . Let $\bar{X} = \frac{1}{19}(X_1 + X_2 + \dots + X_{19})$. Let Q_1 be the mean of $\bar{X}$ and Q_2 the standard deviation of $\bar{X}$. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

2. Let $\bar{X}$ be the sample mean of a sample of size 131 drawn from a normal population with mean -25 and standard deviation 1.7 . Let $Q_1 = P(\bar{X} > -25.02)$, $Q_2 = P(\bar{X} < -25.31)$, and $Q_3 = P(-25.31 < \bar{X} < -24.74)$. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

3. Let Q_1 be the mean and Q_2 the standard deviation of the normal distribution that approximates the sampling distribution of the population proportion if the size of the population is 26796 , the sample size is 638 , and the success probability is 0.636 . Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

4. Using the normal distribution that approximates the sampling distribution of the population proportion if the size of the population is 18684 , the sample size is 519 , and the success probability is 0.876 , let $Q_1 = P(\hat{p} < 0.836)$, $Q_2 = P(\hat{p} > 0.852)$, and $Q_3 = P(0.909 < \hat{p} < 0.919)$, where $\hat{p}$ is the sample proportion. Let $Q = \ln(3 + |Q_1| + 2|Q_2| +$

$3|Q_3|$). Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

5. Let Q_1 be the lower and Q_2 the upper end point of the 93 percent confidence interval for the sample proportion $\hat{p}$ if the success count in the sample is $x = 144$ and $n = 516$ is the sample size. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

6. Let Q_1 be the smallest sample size if we want to estimate a population proportion within 0.09 using a 90 percent confidence interval. Let $Q = \ln(3 + |Q_1|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

7. Relative to a normal population, let Q_1 be the lower and Q_2 the upper end point of the 91 percent confidence interval for the sample mean $\bar{x}$ if the sample mean is $\bar{x} = -94$, the sample standard deviation is $s = 5.9$, and the sample size is $n = 20$. Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

8. Given the numbers below drawn from a normal distribution, let Q_1 be the lower and Q_2 the upper end point of the 94 percent confidence interval for the sample mean.

93.01, 90.89, 92.08, 92.09, 91.81, 92.61, 91.89, 94.22, 92.66, 92.87, 91.88, 93.04, 91.44, 92.34, 90.57

Let $Q = \ln(3 + |Q_1| + 2|Q_2|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

9. Test at the 91 percent level of significance the null hypothesis $H_0: p = 0.572$ versus the alternative hypothesis $H_1: p > 0.572$, where p is the population proportion, $n = 564$ is the sample size, and $x = 340$ is the number of observed “successes”. Let $Q_1 = \hat{p}$ be the sample proportion, Q_2 the z -statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0 , and $Q_3 = 0$ otherwise. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

10. Test at the 91 percent level of significance the null hypothesis $H_0: p = 0.429$ versus the alternative hypothesis $H_1: p \neq 0.429$, where p is the population proportion, $n = 796$ is the sample size, and $x = 381$ is the number of observed “successes”. Let $Q_1 = \hat{p}$ be the sample proportion, Q_2 the z -statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0 , and $Q_3 = 0$ otherwise. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

11. Relative to a normal population, test at the 95 percent level of significance the null hypothesis $H_0: \mu = 363$ versus the alternative hypothesis $H_1: \mu < 363$, where μ is the population mean, $n = 16$ is the sample size, $\bar{x} = 354.27$ is the sample mean, and $s = 8.15$ is the sample standard deviation. Let Q_1 be the degrees of freedom for the t -distribution, Q_2 the t -statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0 , and $Q_3 = 0$ otherwise. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$.

— (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.

12. Relative to a normal population, test at the 95 percent level of significance the null hypothesis $H_0: \mu = 470$ versus the alternative hypothesis $H_1: \mu \neq 470$, where μ is the population mean, $n = 10$ is the sample size, $\bar{x} = 467.85$ is the sample mean, and $s = 2.99$ the sample standard deviation. Let Q_1 be the degrees of freedom for the t -distribution, Q_2 the t -statistic, and $Q_3 = 1$ if we reject the null hypothesis H_0 , and $Q_3 = 0$ otherwise. Let $Q = \ln(3 + |Q_1| + 2|Q_2| + 3|Q_3|)$. Then $T = 5 \sin^2(100Q)$ satisfies:— (A) $0 \leq T < 1$. — (B) $1 \leq T < 2$. — (C) $2 \leq T < 3$. — (D) $3 \leq T < 4$. — (E) $4 \leq T \leq 5$.